

Proof-of-work protocols for AI-generated solutions in mathematics

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Extended Abstract

Proof-of-work protocols can produce for a massive cloud computation an additional certificate for the output, which allows the verification of the output with much less compute effort and possibly at a later time. The soundness of the protocols prevents the prover from giving a false output. Such protocols should be deployable when the prover is an AI-based solver with possibly incorrect substeps.

We discuss using AI-generated proofs by the example of computing the determinant of a very large, dense $n \times n$ matrix. For purpose of simplifying the discussion, we shall assume that the entries are constant integers, whose bit-size does not grow with the dimension n . We note that the discussed protocols could also account for the bit-lengths in the entries. The best proof-of-work protocol for the integer determinant with an $n \text{ polylog}(n)$ bit-sized proof and which is checkable in $n^2 \text{ polylog}(n)$ bit-complexity uses interactivity [4] and Figure 1 below. Here $\text{polylog}(n)$ stands for a function of order $O(\log(n)^k)$ for a constant exponent k . Removal of the interactions is possible by the cryptography-based Fiat-Shamir method [5].

Figure 1 gives the proof-of-work protocol, which is adapted from the first protocol in [4, Section 2] and [2, Section 4.4]. In Step 2 in Figure 1, the bound $2^{\text{polylog}(n)}$ can be determined explicitly from the size bounds of the integers in $\mathcal{A}$ and the probability that $\Delta = \det(\mathcal{A})$ after all verifier checks succeed. The multiplicative pre-conditioner $\Gamma(t, s)$ guarantees that the Prover can achieve the relatively prime characteristic polynomials in Figure 1, Steps 5 and 6: $\text{GCD}(c^B(\lambda), c^C(\lambda)) = 1$. Relative primeness is necessary for the soundness of the checks in Figure 1, Steps 8.

The proof for $\Delta = \det(\mathcal{A})$ in Figure 1, which are the prime p , the residues t, s, r_1 , the polynomials $c^B(\lambda), c^C(\lambda)$ and the vector w , have a combined bit-size of $n \text{ polylog}(n)$. The protocol is high-level and does not specify how the Prover computes the integer Δ , the modular polynomials $c^B(\lambda), c^C(\lambda)$ and the modular linear system solution vector w . Here we suppose that that output is computed by an AI-based solver. The protocol essentially detects incorrect outputs, as it would if the Prover had cheated and not computed the output properly.

A classical method for removing interaction in proof-of-work protocols is in [5]; see also [1]. For Fiat-Shamir'ing, the Prover also selects the random values for p and r_1 in the protocol of Figure 1, but after Δ and after the characteristic polynomials $c^B(\lambda), c^C(\lambda)$ have been computed. One possibility is to use hash values $p = H_1(\mathcal{A}, \Delta)$ and $r_1 = H_2(\mathcal{A}, \Delta, p, c^B, c^C)$, where H_1 and H_2 are cryptographically strong hash functions which the Verifier also can compute. For example, H_1 can be random bits of a cryptographically strong random bits generator, which is seeded with $\mathcal{A}, \Delta$. The unpredictability of the hashed values/cryptographically strong random bits prevents the Prover, for example, of committing a false Δ value in Step 1 in Figure 1, which still satisfies $\Delta \equiv \det(\mathcal{A}) \pmod{p}$. We note that iteration of the Fiat-Shamir method can lead to insecurity [6].

We do not know a non-interactive protocol for the integer determinant which is of $n \text{ polylog}(n)$ bit-size and which has an $n^2 \text{ polylog}(n)$ binary verification complexity. But we note that we have given an $n^2 \text{ polylog}(n)$ bit-sized non-interactive proof in [11]. The proof traces Arne Storjohann's

Fig. 1. A simple integer determinant proof-of-work protocol

$\mathcal{A} \in \mathbb{Z}^{n \times n}$ public, $\det(\mathcal{A}) \neq 0$, $\Gamma(\sigma, \tau) = \begin{bmatrix} \tau & -1 & 0 & \dots & 0 \\ 0 & \tau & -1 & \ddots & \vdots \\ \vdots & 0 & \ddots & \ddots & 0 \\ 0 & \ddots & \tau & -1 & \\ \sigma & 0 & \dots & 0 & \tau \end{bmatrix} \in \mathbb{Z}[\sigma, \tau]^{n \times n}$		
<i>Prover</i>	<i>Communication</i>	<i>Verifier</i>
1. $\Delta = \det(\mathcal{A}) \in \mathbb{Z}$	$\xrightarrow{\Delta}$	
2.	$\xleftarrow{p}$ random prime p with $p \leq 2^{\text{polylog}(n)}$	
3. $A = (\mathcal{A} \bmod p) \in \mathbb{Z}_p^{n \times n}$		
<i>Steps 4–8 verify $\det(A) \in \mathbb{Z}_p$ and $\Delta \equiv \det(A) \pmod{p}$</i>		
4. $B \equiv A \Gamma(t, s)$, $t, s \in \mathbb{Z}_p$ with $t^n + s \not\equiv 0 \pmod{p}$	$\xrightarrow{t, s}$ Checks $t^n + s \equiv \det(\Gamma(t, s)) \not\equiv 0 \pmod{p}$	
5. $c^B(\lambda) = \det(\lambda I_n - B) \in \mathbb{Z}_p[\lambda]$	$\xrightarrow{c^B}$ Note $c^B(0) \equiv (-1)^n \det(A \Gamma(t, s)) \pmod{p}$	
6. $C = [b_{i,j}]_{1 \leq i,j \leq n-1} \in \mathbb{Z}_p^{(n-1) \times (n-1)}$, $c^C(\lambda) = \det(\lambda I_{n-1} - C) \in \mathbb{Z}_p[\lambda]$, $t, s \in \mathbb{Z}_p$ also with $\text{GCD}(c^B, c^C) = 1$	$\xrightarrow{c^C}$ Checks $\text{GCD}(c^B(\lambda), c^C(\lambda)) = 1$ in $\mathbb{Z}_p[\lambda]$	
7.	$\xleftarrow{r_1}$ $r_1 \in \mathbb{Z}_p$ random with $c^B(r_1) \not\equiv 0 \pmod{p}$	
8. Computes $w \in \mathbb{Z}_p^n$ such that		
$(r_1 I_n - B)w \equiv e_n \equiv \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$	$\xrightarrow{w}$ Checks $(r_1 I_n - B)w \equiv e_n \pmod{p}$ and $w_n \equiv \frac{c^C(r_1)}{c^B(r_1)} \pmod{p}$ and $(-1)^n \frac{c^B(0)}{t^n + s} \equiv \Delta \pmod{p}$	

Las Vegas randomized determinant algorithm [12]. Instead of doing the matrix multiplications, the verifier performs a Freivalds check of the products; see [9, Section 2.2] for Freivalds checking. Here, the prover only can use an AI-based matrix multiplication algorithm, which is much finer grain, and the proof is quadratic size.

We propose that AI-based provers should incorporate the Fiat-Shamir paradigm. Instead of cryptographically strong hash functions/random bits generators, one may use a random bits server in the cloud. The AI-based prover then commits parts of the proof to the cloud before receiving random bits. The Verifier then can check the time-stamps of each step in the interaction between the AI solver and the cloud server. With cloud Fiat-Shamir’ing services, interactive proof-of-work protocols such as the GKR protocol [7,8] or linear algebra protocols [2,3] can be deployed with AI-based solvers. In [2] we give an interactive protocol to verify a sum-of-squares proof [10] without fully computing the sum-of-squares.

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